

NURSERY LINK UGANDA

A Web-GIS Marketplace and Supply-Gap Platform for Uganda's Tree Nursery Sector

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APPROVALS

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1. Problem Statement

1.1 The Current Manual Process

Tree nursery information across Uganda, including Mukono District, is currently trapped in scattered spreadsheets and paper logbooks that do not talk to each other:

- i. Non-spatial records: local forestry offices track nursery locations, species stock, and operator contacts on paper or in static, non-georeferenced PDF registries.
- ii. Manual supplier discovery: restoration planners and carbon-project developers cannot easily locate certified seedling suppliers, and must rely on word of mouth or outdated printed lists.
- iii. No livestock visibility: because there is no shared spatial map, buyers cannot check seedling availability, species diversity, or production trends before travelling to a nursery.
- iv. No gap analysis: without spatial overlay tools, no platform in Uganda currently models travel-distance barriers between nurseries and degraded land, so heavily deforested 'Nursery Shadow' areas remain invisible to funders and planners.
- v. Fragmented subsidy access: free seedling giveaways run by NGOs, government, and corporate sponsors are announced through disconnected channels, making it hard for eligible communities to discover and apply for them.

1.2 The Automated Target State

Nursery Link Uganda replaces this fragmented, paper-based landscape with a single, coordinate-accurate Web-GIS platform:

- i. Digital Nursery Atlas: every active nursery is mapped with an exact coordinate, contact details, certification status, and a live, operator-maintained species inventory.
- ii. Integrated marketplace: buyers can search, filter, and either self-navigate to a nursery or check out an Order & Deliver request with automatic distance-based pricing and mobile money escrow.
- iii. Nursery Shadow analytics: road-network service zones (5 km, 10 km, and 20 km) are automatically overlaid against satellite-derived forest-loss data to flag underserved, heavily deforested areas for restoration funding.
- iv. Equity-focused subsidy directory: free seedling campaigns from government, NGO, and corporate funders are listed in one place with eligibility checklists that protect stock from commercial resale.
- v. Species literacy: a Digital Tree Library gives buyers the botanical detail needed to choose the right species before ordering, cross-linked directly to the map.

2. Alternatives Considered

Each major design decision below was weighed against implementation risk, cost, and the constraint that the platform must remain usable on intermittent rural connectivity. The chosen option in each table is the one carried into the architecture from Section 3 onward.

2.1 Web-GIS Mapping Architecture Approach

Option	Description	Why Not Chosen / Why Chosen
A. Static PDF / desktop GIS registry	Continue publishing nursery lists as static PDF documents or desktop-only GIS project files maintained by a forestry office.	Rejected. Not georeferenced in a way the public can query; no real-time stock updates; excludes mobile users entirely.
B. Custom web map (Leaflet + PostGIS) (CHOSEN)	A React frontend renders an open-source Leaflet map backed by a PostgreSQL/PostGIS spatial database that the team fully controls.	Chosen. No per-seat licensing cost, full control over the nursery, species, and order schema, and straightforward integration with the road-network and forest-loss overlay analysis needed for Nursery Shadow detection.
C. Hosted ArcGIS Online dashboard	Build the public map and dashboards directly on Esri's ArcGIS Online platform, reusing the ArcGIS Survey123 field-collection workflow end-to-end.	Not chosen for the public platform. ArcGIS Online licensing and per-user costs do not fit a public marketplace with an open buyer base; however, ArcGIS Survey123 is retained for field data collection, with captured records synchronised into the PostGIS database.

2.2 Nursery Shadow Detection Technique

Option	Description	Why Not Chosen / Why Chosen
A. Straight-line buffer circles	Draw simple fixed-radius circles around each nursery to approximate its service area.	Rejected. Ignores real road networks and terrain; overstates service coverage in areas with poor roads, understates it where a paved road allows longer practical trips.
B. Road-network service zones overlaid on forest-loss data (CHOSEN)	Calculate real travel-distance service zones (5 km, 10 km, 20 km) along the road network for every nursery, then overlay the union of these zones against the Global Forest Change forest-loss layer to flag deforested areas that fall completely outside every zone.	Chosen. Produces a realistic, defensible picture of where seedling access is actually a barrier, directly matching the gap-analysis methodology developed in the underlying research concept note.
C. Manual GIS analyst review	A human GIS analyst periodically reviews nursery coverage and forest-loss maps and manually annotates underserved areas.	Not chosen as the primary method. Too slow to keep pace with new nursery registrations and updated satellite passes; retained only as an optional manual override for edge cases.

2.3 Payment and Escrow Approach

Option	Description	Why Not Chosen / Why Chosen
A. Cash on delivery only	Buyers pay the nursery or courier in cash at the point of delivery or pickup.	Rejected as the sole method. Offers no dispute protection for either party and no digital record for donor or audit reporting.
B. Direct bank transfer	Buyers transfer funds to the nursery operator's bank account before delivery.	Not chosen. Bank account penetration is far lower than mobile money penetration among the rural buyers and smallholder farmers the platform targets.

Option	Description	Why Not Chosen / Why Chosen
C. Mobile money API with escrow (CHOSEN)	Integrate a Ugandan mobile money API (MTN MoMo / Airtel Money); funds are collected at checkout and held in escrow until the buyer confirms delivery, at which point they are released to the nursery operator.	Chosen. Matches the payment method already familiar to the target user base, and escrow protects both buyer and nursery operator from payment disputes.

2.4 Backend Technology Stack

Option	Description	Why Not Chosen / Why Chosen
A. Python (Django or FastAPI)	Python offers a rich geospatial and data-science ecosystem (GeoDjango, GDAL bindings) well suited to the Nursery Shadow overlay analysis.	Not chosen as the primary stack. Would introduce a second language alongside the React frontend, adding operational overhead for a small team; the required PostGIS overlay queries are achievable directly in SQL from a Node.js service.
B. Go with the Gin framework	A statically typed, compiled backend with strong concurrency support for batch geospatial queries.	Not chosen for the first build. Excellent performance profile, but the team's familiarity and the benefit of sharing TypeScript types between the React frontend and backend favoured staying within the JavaScript ecosystem.
C. Node.js with Express (CHOSEN)	A single JavaScript/TypeScript codebase spans the React frontend, the nursery operator dashboard, and the Express backend, using a PostGIS-aware ORM for spatial queries.	Chosen. Keeps the team on one language across the stack, has mature libraries for GeoJSON handling, mobile money API integration, and SMS gateway integration, and is sufficient for the platform's request volume.

3. High-Level System Architecture

The public web portal, the nursery operator dashboard, the admin GIS console, the backend API, the PostGIS database, and the external mobile money, SMS, and satellite data services interact as described below, with the backend API's core logic organised into the modules summarised in the component table.

3.1 Component Responsibilities

Public Web Portal	Lets any visitor browse the nursery map, species library, free seedling campaigns, and news without an account, and lets authenticated buyers check out an order.
Nursery Operator Dashboard	Lets an authenticated nursery operator register their nursery, and maintain live inventory, pricing, and certification details.
Admin GIS Console	Internal console for content moderation, free seedling campaign management, and running the Nursery Shadow overlay analysis.
Backend API	Node.js / Express service routing requests, computing distances and delivery pricing, orchestrating mobile money and SMS calls, and persisting all spatial and transactional data.

PostgreSQL + PostGIS Database	Single source of truth for nursery, species, inventory, order, and campaign records, and the spatial engine behind distance and overlay queries.
Mobile Money & SMS Gateway	Processes buyer payments into escrow and delivers automated order alerts to nursery operators.
Survey123 & Forest-Change Feed	Supplies field-collected nursery records and the periodically updated satellite forest-loss layer used in Nursery Shadow analysis.

3.2 Actors and Use Case Overview

Nursery Link Uganda serves four primary actors, each mapped to a distinct set of use cases that the platform must support:

- i. Buyer / Restoration Planner: browses the nursery map, searches and filters by species or location, views a nursery profile, follows a self-pickup route, or checks out an Order & Deliver request.
- ii. Nursery Operator: registers a nursery profile, maintains live inventory and pricing, and fulfils incoming orders.
- iii. Funder / NGO / Government Partner: lists and manages a free seedling campaign, defines eligibility rules, and monitors the livestock meter.
- iv. System Administrator: moderates nursery listings, manages campaign approvals, and runs the Nursery Shadow overlay analysis.

4. Data Models and Database Schema

The core domain objects behind the nursery marketplace, inventory, ordering, and free seedling campaign features are modelled in a normalized relational schema, described table by table below.

4.1 Key Database Structure

The system uses a normalized relational schema in PostgreSQL with the PostGIS extension providing spatial types and indexing for every location column.

Table nurseries

Field	Type	Notes
id	uuid	Primary key
name, type	text	Operator type: private, commercial, or community
district, sub_county	text	Administrative boundary reference used by the hierarchical filters
location	geometry(Point)	PostGIS point used for distance and service-zone queries
contact, capacity	text, int	Operator contact details and annual seedling capacity
certification_status	text	Standard-certified, pending, or unverified

Table inventory

Field	Type	Notes
id	uuid	Primary key
nursery_id	uuid FK	References nurseries.id
species_id	uuid FK	References species.id
quantity_available	int	Live stock count, edited from the operator dashboard
unit_price	numeric	Price per seedling in UGX

Table orders

Field	Type	Notes
id	uuid	Primary key
user_id, nursery_id	uuid FK	References users.id and nurseries.id
delivery_type	text	self_pickup or order_and_deliver
delivery_fee	numeric	Computed from live road distance at checkout
status	text	pending_payment, escrow_held, fulfilled, or cancelled

Table free_seedling_campaigns

Field	Type	Notes
id	uuid	Primary key
nursery_id	uuid FK	Pickup point for the campaign
funder_name, funder_type	text	Government, international NGO, private foundation, or corporate sponsor
allocated_stock, remaining_stock	int	Drives the live stock meter shown on the campaign profile
eligibility_rules	jsonb	Structured checklist, e.g. LC1 letter required, land-holding thresholds

5. Backend Processing and Nursery Shadow Logic

5.1 Order & Delivery Processing Engine

The backend function ProcessOrderAndDeliver orchestrates a checkout run:

- i. **Trigger:** the buyer selects species and quantities and requests an Order & Deliver quote.
- ii. **Distance Calculation:** the backend issues a PostGIS query computing the road-network travel distance between the buyer's live coordinates and the nursery's location.
- iii. **Pricing:** the delivery fee is calculated from the live distance using the configured motorcycle courier or truck transport weight-rate table.

- iv. **Payment Orchestration:** on confirmation, the backend calls the mobile money API to collect payment into escrow.
- v. **Notification:** once escrow is confirmed, the backend calls the SMS gateway to alert the nursery operator with the buyer's name, delivery coordinates, and itemised order checklist.
- vi. **Persistence:** the order record is written with status `escrow_held`, and released to fulfilled once the buyer confirms delivery in-app.

5.2 Nursery Shadow Detection Engine

The Nursery Shadow analysis is run by an administrator rather than on every page load, since the underlying satellite dataset only refreshes periodically:

- i. **Service Zone Computation:** for every active nursery, the engine computes 5 km, 10 km, and 20 km road-network travel zones and unions them into a single national coverage layer.
- ii. **Forest-Loss Overlay:** the coverage layer is overlaid against the most recently published Global Forest Change forest-loss raster.
- iii. **Shadow Classification:** any heavily deforested polygon that falls entirely outside the 20 km union is classified and stored as a Nursery Shadow zone, visible on the Admin GIS Console.

5.3 Safety and Fallback Logic

- i. **Result caching:** a completed Nursery Shadow run is cached and reused until either new nurseries are added or a newer forest-loss dataset is published, avoiding redundant heavy spatial computation.
- ii. **Payment timeout and retry:** each mobile money API call is bounded by a request timeout with a small number of retries for transient errors before the order is flagged for manual follow-up.
- iii. **Escrow-pending fallback:** if a payment call ultimately fails, the order is left in status `pending_payment` rather than being marked fulfilled, so the buyer and nursery operator both see an accurate state.
- iv. **Batch isolation:** a failure processing one order does not block checkout for other concurrent buyers.

6. REST API Specifications

The endpoint table below aligns with the primary Express routes in the backend API.

Method	Endpoint	Description	Auth Role
GET	/api/v1/nurseries	Returns nurseries filtered by district, sub-county, species, or search text.	public
GET	/api/v1/nurseries/:id	Returns a single nursery profile with live inventory.	public
POST	/api/v1/orders/quote	Computes delivery distance and fee for a proposed order.	buyer
POST	/api/v1/orders	Submits an order, triggers mobile money escrow and the SMS alert.	buyer
PUT	/api/v1/orders/:id/confirm-delivery	Buyer confirms delivery, releasing escrow to the nursery operator.	buyer

Method	Endpoint	Description	Auth Role
PUT	/api/v1/nurseries/:id/inventory	Nursery operator updates species stock quantities and pricing.	operator
GET	/api/v1/campaigns	Lists active free seedling campaigns filtered by sub-county or purpose.	public
POST	/api/v1/admin/shadow-analysis/run	Triggers a Nursery Shadow overlay run against the latest forest-loss dataset.	admin

6.1 Submit Order, Example Payload Fields

nursery_id	uuid of the fulfilling nursery
items	list of { inventory_id, quantity }
delivery_type	self_pickup or order_and_deliver
delivery_coordinates	buyer's live latitude and longitude, required when delivery_type is order_and_deliver
payment_method	momo or airtel_money

7. Frontend UI/UX Design

The interface follows the platform's established visual language: a deep-green primary palette, clean sans-serif typography, and a split-screen layout on the core Nurseries page, consistent with the project's design blueprint.

7.1 Nurseries Dashboard (Web-GIS Split Screen)

Route: /nurseries.

- Left panel: the interactive Web-GIS map engine, showing clustered pins at national level, boundary highlights at district level, and named pins at sub-county level.
- Right panel: a search box, hierarchical District / Sub-County dropdown filters, a 'Sort by Nearest' toggle, and a scrollable, alphabetically ordered nursery list synced to the current filter.
- Selecting a pin or a list entry opens the nursery pop-up card described in Section 7.2.

7.2 Nursery Pop-up Card and Order & Deliver Panel

Triggered by tapping a map pin or a list entry.

- Header: nursery name, operator type, and certification badge.
- Body: location, annual capacity, seed source, contact details, live distance from the buyer, and stock category icons.
- Inventory list: species names (each cross-linked to its Digital Tree Library entry), quantity available, and unit price.
- Action row: a 'Follow Map to Nursery' button for self-guided pickup, and an 'Order & Deliver' button that expands the quantity-input and mobile money checkout flow described in Section 5.1.

8. Implementation Roadmap

The roadmap below sequences the platform's build across an eight-week schedule, starting from the data model and mapping engine and ending with the Nursery Shadow analytics layer.

Weeks	Scope
1 - 2	Define the requirements, the PostGIS data model, and load district / sub-county boundary and reference species data.
2 - 4	Build the PostGIS database and the Web-GIS mapping engine across national, district, and sub-county zoom tiers.
4 - 6	Implement the nursery marketplace: search, hierarchical filters, the pop-up card, and the Order & Deliver checkout flow.
6 - 7	Integrate the mobile money API and SMS escrow alerts; build the Free Seedlings module and Digital Tree Library.
7 - 8	Implement the Nursery Shadow analytics engine (service-zone and forest-loss overlay) and run end-to-end integration testing.

9. Risks and Assumptions

Item	Detail
Compressed timeline	The eight-week roadmap in Section 8 protects the mapping engine and marketplace checkout as the must-ship core deliverables ahead of the Nursery Shadow analytics layer.
Data quality assumption	Nursery Shadow accuracy depends on nursery operators keeping inventory and location data current; stale records reduce both marketplace and gap-analysis reliability.
Satellite data latency	The Global Forest Change dataset refreshes periodically rather than in real time, so Nursery Shadow zones reflect the most recently published forest-loss layer, not live deforestation.
Mobile money availability	The safety and fallback logic in Section 5.3 leaves an order in pending_payment rather than blocking checkout for other buyers if the mobile money API is slow or unavailable.
Rural connectivity	Intermittent internet access in target districts may affect map tile loading and live GPS sort; the interface should degrade to the last-cached filtered list rather than fail outright.